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Technical Report

Microstructural, physico-chemical and mechanical characterisation of *Sansevieria cylindrica* fibres – An exploratory investigation

V.S. Sreenivasan [a], S. Somasundaram [b], D. Ravindran [c,*], V. Manikandan [d], R. Narayanasamy [e]

[a] Department of Mechanical Engineering, Dr. Sivanthi Aditanar College of Engineering, Tiruchendur 628 215, Tamil Nadu, India
[b] Government Sidha Medical College, Palayamkottai 627 002, Tamil Nadu, India
[c] Department of Mechanical Engineering, National Engineering College, Kovilpatti 628 503, Tamil Nadu, India
[d] S. Veerasamy Chettiyar College of Engineering and Technology, Puliankudi 627 855, Tamil Nadu, India
[e] Department of Production Engineering, National Institute of Technology, Tiruchirapalli 620 015, Tamil Nadu, India

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ABSTRACT

The microstructural, physical, chemical and mechanical properties of *Sansevieria cylindrica* fibres are described for the first time in this work. A microstructural analysis of *S. cylindrica* leaves showed the presence of structural fibres and arch fibres. Polarised light microscopy and scanning electron microscopy of these fibres revealed a hierarchical cell structure that consisted of a primary wall, a secondary wall, a fibre lumen and middle lamellae. The cross-sectional area and porosity fraction of the fibre were estimated to be approximately 0.0245 mm^2 and 37%, respectively. The fibre density and fineness were approximately 0.915 ± 0.005 g/cm^3 and 9 Tex, respectively. An X-ray diffraction and Fourier transform infrared analysis of the fibres showed the presence of cellulose I_β with a crystallinity index of 60%. Tensile tests showed that the corrected Young's modulus was approximately 7 GPa, the tensile strength was 658 MPa, and the total elongation was between 10% and 12%.

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1. Introduction

Environmental awareness groups all over the world have focused their attention on the use of cellulose fibres to reinforce polymer matrixes. The attractive features of natural fibres include their low cost, light-weight, moderate strength, high specific modulus, renewability, biodegradability, lack of health hazards and amenability to chemical modification. Therefore, natural fibre-based composites have good potential for use as building materials. Several authors have reported the use of natural fibres such as palmyra [1], sisal [2], banana [3], oil palm [4], henequen [5], jute [6], hemp [7] and wood pulp [8] as reinforcements in polymer matrixes.

The industrial use of natural fibres as reinforcements in composite materials started at the beginning of the 20th century with the manufacturing of large quantities of sheets, tubes and pipes for electronic purposes. For example, the seats and fuel tanks of aircraft were made of natural fibres with a small content of polymeric binders [9]. When cost-effective synthetic fibres that were less sensitive to temperature and moisture were brought onto the market, natural fibres were largely abandoned in these industries. More recently, stringent environmental regulations and an increased interest in the use of natural resources have led to a positive change among composite industries and end users. Efforts are being made to find alternate reinforcements and resin systems that are eco-friendly and provide the same performance as their synthetic counterparts [10].

Today, a revolution in the use of natural fibres as reinforcements in technical applications is taking place, primarily in the automotive industry. European renewable fibres such as flax and hemp are now used to manufacture door panels and the roofs of automobiles [10]. However, accelerating the substitution of synthetic fibres by natural fibres requires the greater availability of such fibres, and their current production level does not meet today's demand. New plants must be found that enable easy and cost-effective extraction methods that do not impair the properties of the fibre. These new fibres must be analysed to determine their physical, chemical and mechanical properties. Microscopy (either optical or electron) is an invaluable tool to strengthen our knowledge of the morphology of fibres. This knowledge is essential to evaluate or efficiently simulate the properties of these fibres.

In this paper, the *Sansevieria cylindrica* and fibres extracted from it are described. The present study aims to investigate the potential use of *S. cylindrica* fibres (SCFs) as reinforcements in polymeric materials. The physical, chemical and mechanical properties of the SCFs were measured and compared with other natural fibres. A pycnometer was used to assess the density of the fibres, and a

* Corresponding author. Tel.: +91 94432 57441; fax: +91 4632 232749.
E-mail addresses: vssreenivasan@ymail.com (V.S. Sreenivasan), pkssomasundaram@yahoo.in (S. Somasundaram), rmkv1@rediffmail.com (D. Ravindran), vaimanikandan@yahoo.com (V. Manikandan), narayan@nitt.edu (R. Narayanasamy).

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Fig. 1. *Sansevieria cylindrica* with striped, round, green–gray leaves.

chemical analysis was performed to determine their lignin, cellulose, hemicellulose, wax and moisture content. The chemical analysis of the SCF was fine tuned using X-ray diffraction (XRD) and Fourier transform infrared (FTIR) analysis. A microscopic examination was carried out with a polarised light microscope and a scanning electron microscope (SEM). The tensile properties of the fibre were measured using an INSTRON universal testing machine.

2. Natural fibre material

Plants of *S. cylindrica Wenceslas Bojer* [11], belongs to Ruscaceae were collected from Papanasam in Western Ghats, Tamil Nadu, South India, and established in a home garden at Vickramasingapuram in Tamil Nadu, India. This plant was used to obtain the natural fibres.

2.1. S. cylindrica and its leaves

S. cylindrica (Ruscaceae), also known as the Cylindrical Snake Plant or Spear Sansevieria, is a succulent plant native to Angola. *S. cylindrica* grows in a fan shape with stiff leaves growing from the base of the plant in a rosette pattern. Fig. 1 shows a typical specimen of *S. cylindrica*, which is naturally available in India. This plant has striped, round leaves that are smooth and green–gray in color[1]. In Fig. 1, the plant contains eight leaves. A single leaf is about 30 mm thick and grows to a height between 1000 mm and 2000 mm. The *S. cylindrica* leaf consists of a sandwich structure composed of approximately 5% fibre, 1% cuticle, 10% dry matter and 84% water.

S. cylindrica produces 100–150 leaves before flowering. Each of these leaves contains approximately 600–1000 fibre bundles that are approximately 1000–2000 mm long. A *S. cylindrica* leaf yields a strong white elastic fibre, which has long been used by the native inhabitants for mats, hammocks, bowstrings and other types of crude cordage. SCF can be used as a reinforcement in cement and polymer-based composites [12]. The SCFs were removed from the leaf by hand or mechanically. SCFs can be incorporated as short, randomly distributed fibres, long oriented fibres or as a fibre-fabric in matrixes.

[1] For interpretation of color in Figs. 1,3–7,9,11–14, the reader is referred to the web version of this article.

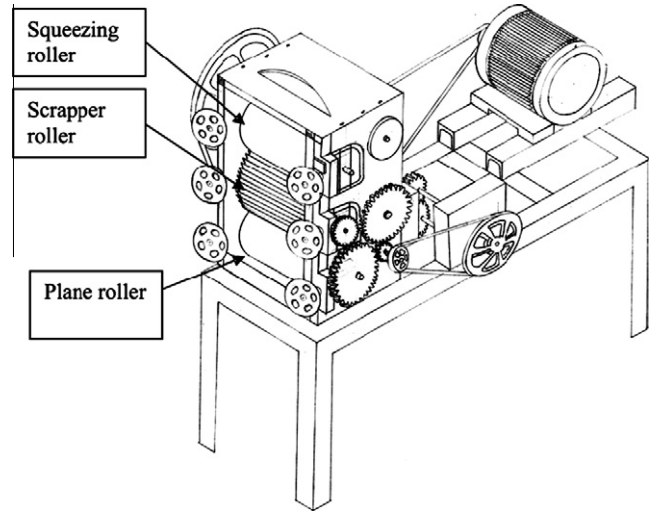

Fig. 2. Schematic diagram of a mechanical decorticator.

2.2. Extraction procedure for SCFs

The SCFs were separated from the *S. cylindrica* leaves, which were collected from farms around the city of Tirunelveli in Tamil Nadu, India, using a mechanical process called decortication. In this process, the *S. cylindrica* leaves were fed into a fibre-extracting machine called a mechanical decorticator. A schematic diagram of a mechanical decorticator is shown in Fig. 2. The machine consists of a pair of feed rollers and a beater. The leaves were fed into the beater through the feed rollers between a squeezing roller and a scrapper roller. The fibres were extracted, and the pulps were separated. The decorticated fibres were dried in the sunlight to remove the moisture content, and machine combed for separation. In this study, the washed SCFs were dried for 24 h in sunlight and then used for characterisation. The separated SCFs are shown in Fig. 3.

3. Experimental procedure

3.1. Microstructural studies of S. cylindrica leaf and fibres

3.1.1. Preparation of specimens and sectioning

Healthy *S. cylindrica* leaf was collected for microstructural analysis. For anatomical studies, the leaf was cut into small pieces (10 mm × 10 mm) and fixed in FAA (5 ml formaldehyde + 5 ml acetic acid + 90 ml 70% ethyl alcohol). After 24 h, the specimens were dehydrated through a graded tertiary-butyl alcohol series and embedded in paraffin [13]. Sections of 10–12 μm were cut on a rotary microtome, affixed to a glass slide and stained with a mixture of Toluidine blue, safranine, fast green and Lugo's iodine solution [14]. Polarised light micrographs were taken with a Nikon Labphoto 2 microscopic unit. For normal observations, bright field imaging was used. For the crystal (starch grains and lignified cells) studies, polarised light was employed.

3.1.2. Scanning electron microscopy

Transverse sections of *S. cylindrica* leaf and fibres were examined using a scanning electron microscope, HITACHI Model S3000H. The leaf section and fibres were given uniform gold coating on all sides except their cross-sections to make the surfaces conductive. The cross-sections of the *S. cylindrica* leaf and fibres

Fig. 3. Dried SCFs are strong and white.

were examined at different magnifications. In addition, the longitudinal view of the fibre was also observed.

3.2. Physical analysis of SCFs

The goal of this section is to describe the methodology for measuring the fibre cross-section, its microstructural features and the physical properties of fibre (i.e., its density and fineness).

Optical microscopy was used to investigate the shape of the fibre. The fibre cross-sections were roughly irregular. To measure the cross-sectional area of the fibre, an adjacent piece of tensile tested fibre (the piece close to the broken portion) was examined using the SEM. To evaluate the variability of the cross-sectional area through the length, a 40-mm-long fibre was cut into four sections of 10 mm each, and four micrographs were taken. The images were post-processed using ImageJ. A contour line was drawn to delineate the fibre cross-section (Fig. 4), and the area was determined.

The dimensions of various microscopic features (i.e., the middle lamellae, primary wall, secondary wall and fibre lumen) of the SCF were calculated using the SEM and ImageJ. Moreover, the porosity fraction of the SCF cross-section was computed using SEM and ImageJ. The relation between the real section (needed to compute the stresses efficiently) and the raw section was then established.

The density measurement of the SCF was carried out using a pycnometer for solids with toluene as the immersion liquid following the procedure of Beakou et al. [10]. The fibres were dried for 96 h in a desiccator containing silica. They were then cut into lengths of 5–15 mm and introduced into the pycnometer, which was then placed in the desiccator for 24 h. The temperature in the room was 18.1 °C. The hygrometry was 57% in the room and only 3% in the desiccator. Before carrying out the hydrostatic weighing with toluene, the fibres were impregnated in toluene for 2 h to evacuate the micro bubbles in the fibres. The density of toluene is $\rho_t = 0.866$ g/cm^3.

The density of the SCF was calculated using the following formula:

$$\rho_{SCF} = \frac{(m_2 - m_1)}{(m_3 - m_1)(m_4 - m_2)}\rho_t \quad (1)$$

where ρ_{SCF} is the density of the SCF (g/cm^3), ρ_t is the density of the toluene (g/cm^3), m_1 is the mass of the empty pycnometer (kg), m_2 is the mass of the pycnometer filled with chopped fibres (kg), m_3 is the mass of the pycnometer filled with toluene (kg) and m_4 is the mass of the pycnometer filled with chopped fibres and toluene (kg).

The fineness of the SCF was determined in terms of linear density in accordance with ASTM D 1577-92. The linear density was obtained as Tex and *denier* from the weight of 100 single fibres that were each 60 mm in length.

3.3. Chemical analysis of the SCFs

The goal of this analysis was to determine the content of lignin, cellulose, hemicelluloses, wax and moisture of the SCF using conventional chemical, XRD and FTIR analyses.

The determination of the lignin content was carried out according to the Klason method [15]. The samples were crushed and extracted with dichloromethane before being hydrolysed in a 72% solution of sulphuric acid. Lignin, which was the only insoluble component, was separated from the fibre and quantified. The cel-

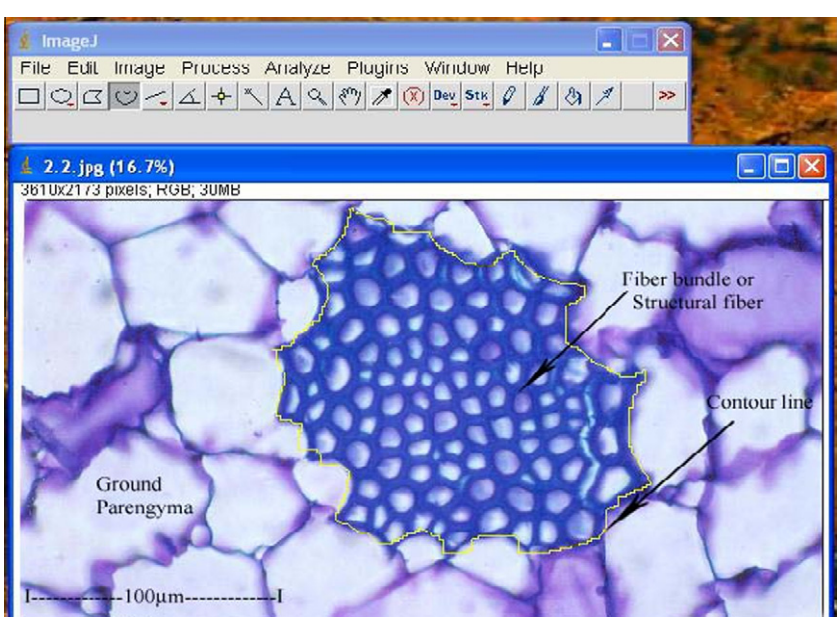

Fig. 4. Area calculation using ImageJ from the polarised light micrograph showing a transverse section of the structural fibre (40×).

lulose content was measured according to Kurshner and Hoffer's method [16]. The samples were crushed and extracted with dichloromethane, and then a mixture of ethanol and 95% nitric acid was added. The cellulose that corresponded to the insoluble fraction of the samples was weighed. The determination of hemicellulose was carried out according to standard NFT 12-008. The samples were heated in hydrobromic acid. The hemicellulose was transformed into furfural, and the latter was extracted by distillation and measured by spectrophotometry.

The wax content in the SCF was determined following the method developed by Conrad [17].The SCF samples were crushed and subjected to soxlet extraction using ethanol for 6 h. The resulting solution contained sugar, wax and other alcohol-soluble substances. The solution was transferred to a separator funnel, and chloroform was added to extract the wax from the alcohol solution. Purified water was then added, and two separate layers of chloroform and alcohol were formed. After separation, the chloroform evaporated from the solution, leaving a waxy residue. After extraction, the dried residue and the SCF samples were weighed, and the wax content was expressed in terms of the original weight of the SCF samples.

The moisture content in the SCF was determined by the following procedure. Weighed SCF samples were dried in an oven at 104 °C for 4 h, and the dried samples were weighed. The moisture content in the SCF was determined using the following expression:

$$\%\text{of moisture content in SCF} = \frac{\{\text{Sample weight before drying}\} - \{\text{Sample weight after drying}\}}{\text{Sample weight before drying}} \times 100 \quad (2)$$

X-ray spectra (scan range (2θ) = 10–80°; θ = diffraction angle; scan speed = 5.0 deg/min) of the SCF were obtained with a Rigaku X-ray diffractometer D/Max Ultima III with an X-ray tube producing monochromatic Cu Kα radiation. The integrated intensities of the Bragg peaks in the spectrum of the SCF were identified, and their crystallinity indices were estimated.

FTIR spectra of the SCF were recorded using a Perkin Elmer Spectrum RXI FTIR spectrometer in a KBr matrix with a scan rate of 32 scans per minute and a resolution of 2 cm^{-1} in the wave number region from 400–4000 cm^{-1}. The SCF samples were chopped into small particles using scissors and ground to a fine powder using a mortar and pestle. This powder was mixed with KBr and pelletised by pressurisation to record the FTIR spectra under standard conditions.

3.4. Tensile behaviour of the SCFs

The dried SCFs were tested in dry conditions under tensile loading at gauge lengths (GL) of 10, 20, 30 and 40 mm in an INSTRON universal testing machine of type 5500 R (Fig. 5) according to the ASTM D 3822-01 standard. The GL was varied to determine its effect on the tensile properties. Pneumatic grips were used to clamp the fibre with a pressure of 0.4 MPa. A 1.0 kN capacity load cell was used to measure the load. The displacement of the fibre was measured by a short-stroke transducer with a resolution of approximately 0.1 μm. The tensile tests were conducted with a cross head speed of 0.1 mm/min. The average strain rates were on the order of 0.6 s^{-1} and 0.15 s^{-1} for the gauge lengths of 10 mm and 40 mm, respectively. Due to the variability of the natural fibres, 20 samples were tested at each GL, and the average value was reported. All testing was conducted at ambient temperature (~21 °C) and a relative humidity of about 65%.

The compliance of the loading and gripping systems was determined by obtaining the force versus displacement behaviour of the fibre at various gauge lengths following the methodology used by Chawla et al. [18]. The total cross–head displacement during fibre testing, δt, is expressed using the following expression:

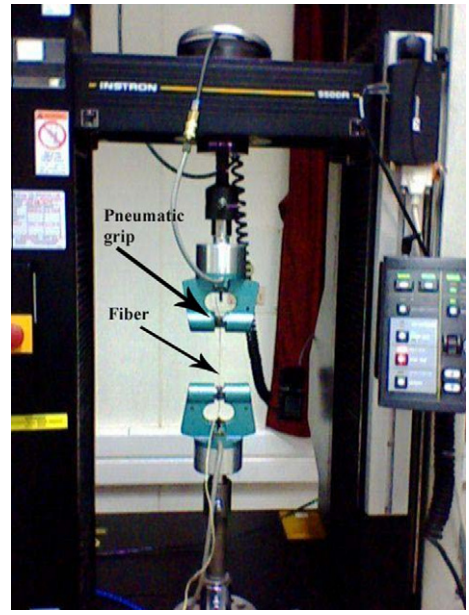

Fig. 5. INSTRON universal testing machine used for tensile testing.

$$\delta t/F = (1/\text{EA}) + c \quad (3)$$

where F is the applied force, l is the GL, E is the Young's modulus of the fibre, A is the cross-sectional area of the fibre and c is the machine compliance.

A plot of $\delta t/F$ versus GL (l) yields a straight line of Slope 1/(EA) and intercept c, which is the compliance of the load train.

4. Results and discussion

4.1. Microstructural analysis of S. cylindrica leaf and fibres

In the transverse sections, the leaf displayed a dermal tissue system, a ground tissue system and a vascular tissue system.

The dermal tissue consisted of a well-defined epidermis with radially oblong, fairly wide cells that have thick cuticles (see Fig. 6a). The epidermal layer had cuticles that were 70 μm thick. The ground tissue was homogeneous and parenchymatous. The cells were circular to polygonal, thin walled and compact. Fibre bundles were observed in the ground tissue (Fig. 6a). The vascular bundles were mostly located in the central portion of the ground tissue. The vascular bundles had two lateral rows of wide, angular, thick-walled xylem elements and three or four small nests of phloem elements situated along the outer part of the xylem strands (Fig. 6b). The xylem elements were 50 μm wide, and their walls were 5 μm thick. The lumen of the xylem elements was 40 μm. Fibres were also present at one end of the vascular bundle.

Within the leaf, there were two basic types of fibres: a structural fibre or fibre bundle and an arch fibre or bundle-cap fibre. The structural fibres provide stiffness for the *S. cylindrica* leaf and were found in the periphery of the leaf (see Fig. 6a). The cross-section of the structural fibre was polyhedral in outline and compact (Fig. 6c). Fig. 7 shows an SEM image of the transverse section of the structural fibre. In this image, a flaky cell structure with small voids is observed. The equivalent diameter of the structural fibres was in the range of 50–100 μm. The structural fibres were surrounded by ordinary ground parenchyma cells. The structural fi-

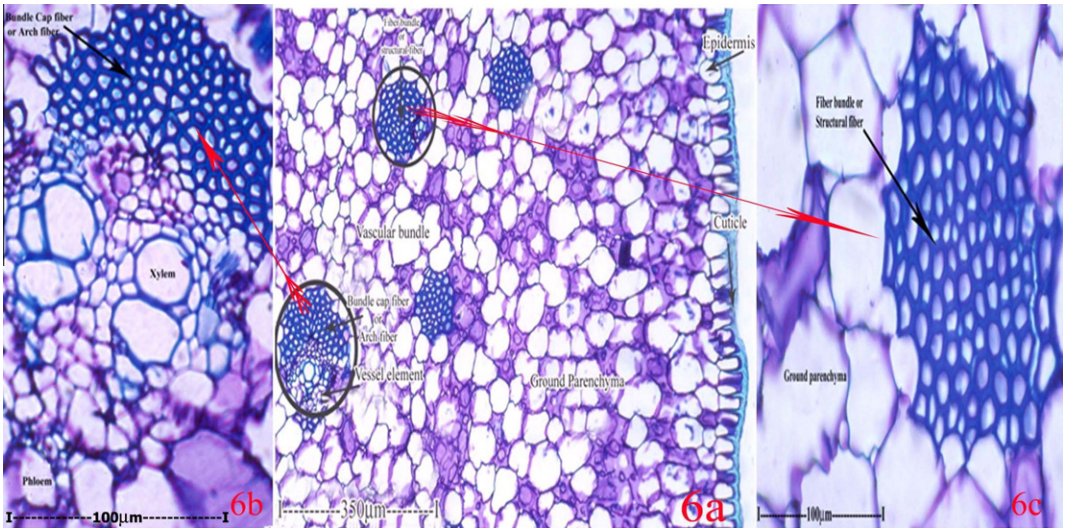

Fig. 6. (a) Transverse section of a *Sansevieria cylindrica* leaf showing fibre bundles and vascular bundles. (polarised light micrograph; 10×). (b) An enlarged vascular bundle showing two parallel rows of xylem elements, four phloem strands along the periphery and a thick bundle-cap fibre (40×). (c) Magnified view of a transverse section of a structural fibre (40×).

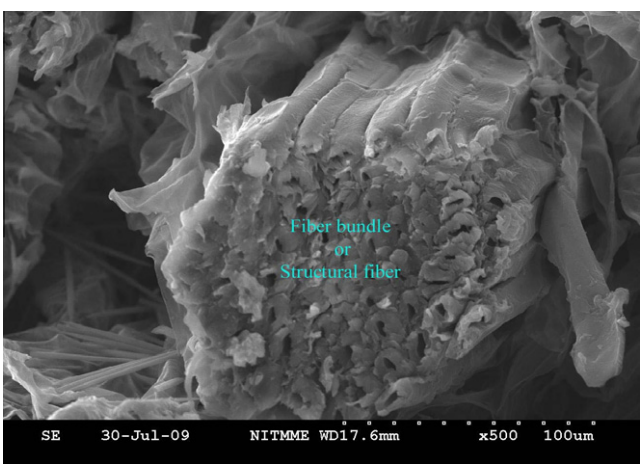

Fig. 7. SEM view of a transverse section of a structural fibre.

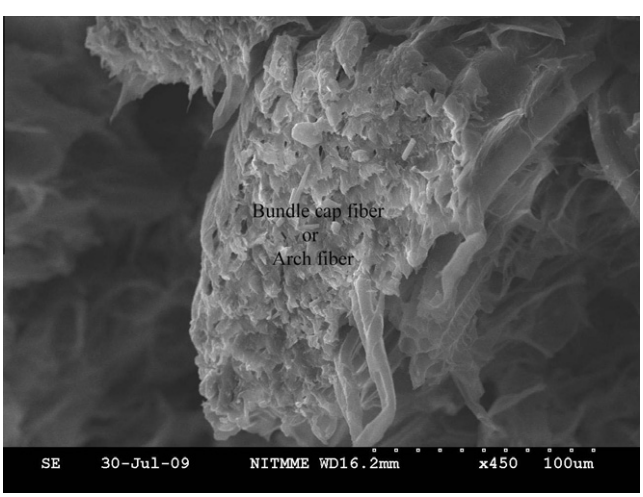

Fig. 8. SEM view of a transverse section of an arch fibre.

bres are of great importance commercially because they almost never split during the extraction process.

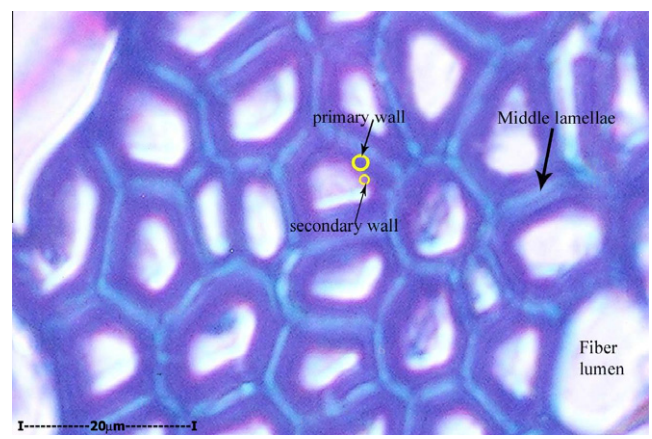

Fig. 9. Transverse section of fibres showing primary walls, secondary walls and lumens (polarised light micrograph; 60×).

The arch fibres grow at one end of the vascular bundle and were usually found in the middle of the leaf (Fig. 6a). The cross-sectional view of the arch fibres shows a semi-elongated circular shape (Figs. 6b and 8). These fibres run from the base to the tip of the plant. The structural fibres were less lignified than the arch fibres.

The SCFs had a hierarchical structure. Each fibre contained numerous elongated individual fibre cells that were approximately 6–30 µm in diameter. Each fibre cell is made up of three main parts: the primary wall, the secondary wall and the fibre lumen (Fig. 9). The fibre cell walls were heavily lignified and responsible for fibre strength. The primary wall contained cellulose, and the secondary wall consisted of lignin. The fibre lumen was approximately 6–10 µm wide. The fibre cells were linked together by the middle lamellae, which consisted of hemicellulose and lignin. Fig. 10 shows a longitudinal view of the SCF. The longitudinal surface of the SCF resembled a combination of the linear fibre segments. The longitudinal surface of the SCF also consisted of several surface irregularities that provide good interfaces between the fibre and the resin when SCFs are used to manufacture composites.

Our results are in according with the findings of Murherjee et al. [19] and Nutman et al. [20]. Both groups conducted a microstructural analysis of various types of sisal fibres and fibre cells. They

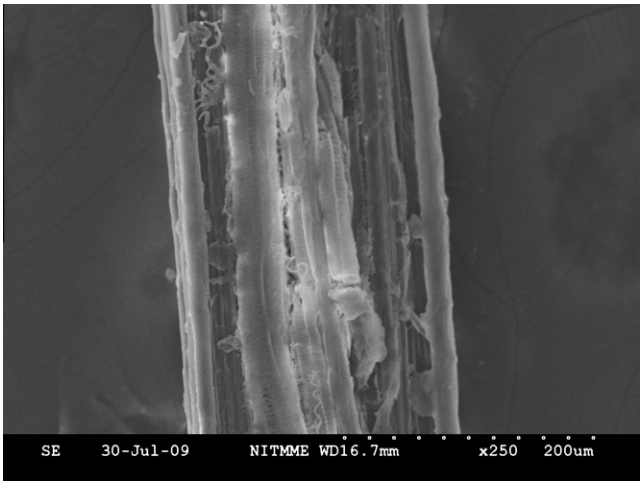

Fig. 10. SEM image of a longitudinal view of SCF.

observed the presence of structural, arch and xylem fibres in sisal leaves. Beakou et al. [10] analysed the microstructure of a *Rhectophyllum camerunense* plant fibre. They observed that *R. camerunense* fibres extracted from the air roots of a plant contain primary walls, secondary walls and lumen.

4.2. Physical and chemical analysis of SCF

Fig. 6b and c shows that the fibre cross-section is irregular. Because SCFs have irregular cross-sections, simply measuring the diameter at a given location may result in an erroneous measurement of the cross-sectional area of the fibre. Image analysis provides a very precise method for measuring the area. An area measurement at four randomly selected, equally spaced points on a 40 mm SCF sample resulted in values of 0.023, 0.028, 0.024, and 0.023 mm^2. The average value of the area was calculated and reported.

The porosity fraction in the fibres was determined by image processing using ImageJ. The porosity fraction of the fibres is defined as the ratio of the total surface area of the fibre lumens to the raw surface area of the fibre cross-section [10]. In this study, the porosity fraction was calculated using a sample of 10 sections, and the results indicated that the porosity fraction varied from 32% to 37%. This high porosity fraction promotes good impregnation when a low viscosity resin is used for composite fabrication. The same analysis also allowed us to evaluate the various feature dimensions in the microscopic structures of SCFs using the image shown in Fig. 9. These dimensions are reported in Table 1.

Table 1
Average thickness of various features in the microstructure of SCFs.

Fiber microstructural features	Middle lamellae	Primary wall	Secondary wall	Fiber lumen
Thickness average (nm)	1112 ± 55	1558 ± 82	667 ± 28	8889 ± 424

The data in Table 1 indicate that the primary wall is the thickest layer. Its thickness was approximately triple that of the secondary wall. The cellulose content of the primary wall was also the highest of all layers in the fibre cell.

As expected by the high porosity fraction, the density of the SCF computed using Eq. (1) was 0.915 ± 0.005 g/cm^3. This low density (Table 2) is favourable for light-weight applications. Another application for this low density and high porosity fraction is in the construction of thermal and acoustic insulation using panels reinforced with SCF. The fineness of SCF was determined in terms of the linear density, which was 8.95 Tex or 80.52 *denier*.

The density and chemical composition of SCFs and the reported values of other natural fibres are shown in Table 2. The chemical composition of the fibre influences its properties and, in many ways, the fibres themselves can be considered fibrous composite materials [23]. For example, the cellulose content has an important influence on the mechanical properties of natural fibres [24]. As shown in Table 2, SCFs contain 79.7% cellulose, which is ~85% greater than that of coir fibre and 2% less than that of pineapple fibre.

SCFs also contain 10.13% hemicellulose, 3.8% lignin, 0.09% wax and 6.08% moisture. The hemicellulose content in SCFs is less than other natural fibres. The lignin content influences the fibre structure, properties and morphology [23], and the wax content in raw fibres affects the interfacial bond between the fibre and matrix during composite formation. However, SCFs contain only 0.09% wax, which is a negligible quantity. Therefore, the interfacial bond between raw SCFs and the resin is not affected by wax.

4.2.1. XRD analysis

Fig. 11 displays a typical X-ray spectrum for SCFs. This spectrum contains two peaks, which are particularly well-defined for a raw fibre. The presence of these diffraction peaks indicates that the fibre is semi-crystalline. According to several authors [25,26], the two peaks situated at $2\theta = 15.4°$ and $2\theta = 22.5°$ can be attributed to cellulose I and IV, both of which exhibit a monoclinic structure [25–27]. However, the FTIR analysis also showed a characteristic band for cellulose I_β [26–28]. This result indicated that the type of cellulose present in the fibre is cellulose I_β.

These two peaks are attributed to the (2 0 0) and (1 1 0) crystallographic planes [25,29,30]. The crystallinity index (CI) was estimated using the following expression [31]:

$$CI = \frac{H_{22.5} - H_{18.5}}{H_{22.5}} \quad (4)$$

where $H_{22.5}$ is the height of the peak at $2\theta = 22.5°$. This is due to the contribution of both the amorphous and the crystalline fractions.

Table 2
Chemical composition and density of SCFs and different cellulose-based natural fibres.

Fiber	Cellulose (%)	Hemicelluloses (%)	Lignin (%)	Wax (%)	Moisture (%)	Density (g/cm^3)	References
SCF	79.7	10.13	3.8	0.09	6.08	0.915	–
Banana	63–64	19	5	–	10–11	1.35	[21,22]
Coir	43	–	45	–	–	1.15	[22]
Sisal	67	12	12	–	–	1.45	[9,22]
Pineapple	81	–	12	–	–	1.44	[22]
Jute	64.4	12	11.8	–	–	1.3	[9]
Flax	64.1	16.7	2	–	–	1.5	[9]
Ramie	68.6	13.1	0.6	–	–	–	[9]
Rhectophyllum camerunense	68.2	16	15.6	–	–	0.947	[10]

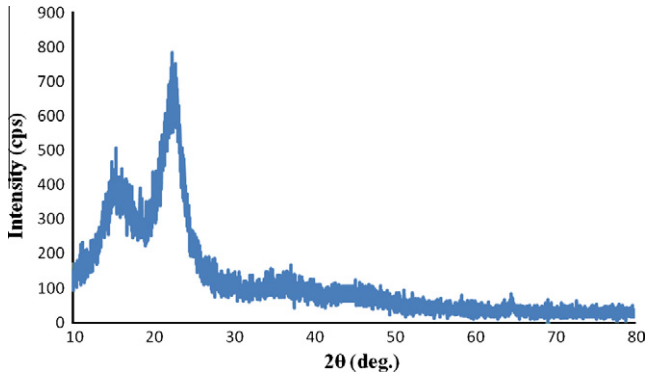

Fig. 11. X-ray spectrum of SCF.

$H_{18.5}$ is the diffracted intensity at $2\theta = 18.5°$ and is attributed to the amorphous fraction [30,32]. This calculation yields a CI value of 60%. This value is higher than that of the *Wrighitia tinctoria* seed fibre (49.2%) and ramie (58%); it is equal to the value for cotton (60%) and smaller than the value for *raffia textilis* (64%), sisal (71%), jute (71%), flax (80%) and hemp (88%) [26,30–32]. This result suggests that the crystallites of SCF are less ordered than those of raffia textilis, sisal, flax and hemp. An estimate of crystallite size (L) was performed using Scherrer's formula [31]:

$$L = \frac{K\lambda}{\beta \cos\theta} \tag{5}$$

where $K = 0.89$ is Scherrer's constant, β is the peak's full-width at half-maximum and λ is the wavelength of the radiation [25,27,33]. When Eq. (5) was applied to the (2 0 0) peak, the crystallite size (L) was 86 nm. This value is more than twice that of the crystallite size of raffia textilis fibres ($L = 32$ nm) calculated using the same framework [31], and it is far larger than that of ramie fibres (16 nm) [34], cotton fibres (5.5 nm), cornstalk fibres (3.8 nm) and flax fibres (2.8 nm), as reported by Reddy and Yang [29]. This large crystallite size ($L = 86$ nm) reduces the chemical reactivity and water absorption capacity of the material.

4.2.2. FTIR analysis

The FTIR spectra for SCF are presented in Fig. 12. The reported values of the band positions vary from author to author [27,28,32,35,36]. The positions of the vegetable fibre bands vary by about 16 cm^{-1}, and a given band can have several origins [36]. Therefore, the band positions vary between studies. When the variability of the position is taken into account, the bands at 3400 cm^{-1} (3407.33 in Fig. 12) and 760 cm^{-1} (771.67 in Fig. 12) can be attributed to cellulose I_β [27,28]. The bands at 1740 cm^{-1} (1730 in Fig. 12) and 1510 cm^{-1} (1509.15 in Fig. 12) are attributed to lignin [27].

The CH stretch at 2936.51 cm^{-1} is present in the spectrum. The carbonyl band at 1730 cm^{-1} can be seen in the spectrum. The band at 1730 cm^{-1} is also attributed to the C=O stretch of the acetyl groups of hemicelluloses [37,38]. The COOH bending band and the CH_2 symmetric bending band were present at 662.41 cm^{-1} and 1361.87 cm^{-1}, respectively. The band at 1260.31 cm^{-1} is the C—O stretch of the acetyl group of lignin [38]. The C=O bonds on the hemicelluloses are represented by the band 1612.85 cm^{-1} [30]. The band at 771.67 cm^{-1} represents the presence of saline [39].

4.3. Tensile behaviour of SCF

A total of 20 fibres were randomly chosen from a given batch and tested for each GL. The Young's modulus was calculated in the elastic portion of the stress–strain curve and then corrected for compliance by measuring the force versus the displacement at various gauge lengths using Eq. (3) (Fig. 13). These results are shown in Table 3. The GL does not influence the modulus of the fibre. The variability in the modulus for a given GL is due to the variability in the microstructure of *S. cylindrica* fibres and also due to possible damage that occurred during the extraction process.

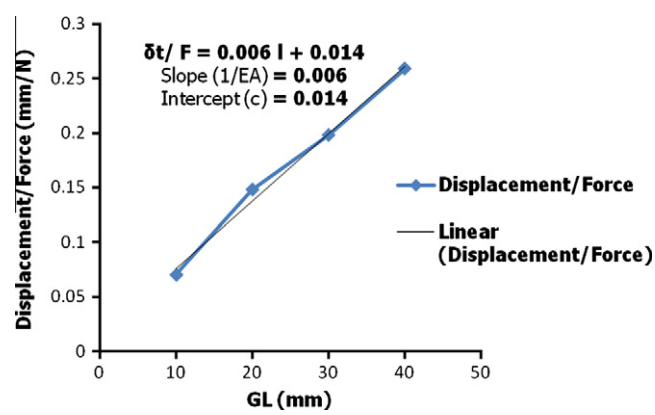

Fig. 13. Normalised displacement versus GL plot showing a linear relationship with a slope of 1/(EA). The machine compliance, which is given by the intercept ($c = 0.014$), was also determined from this plot.

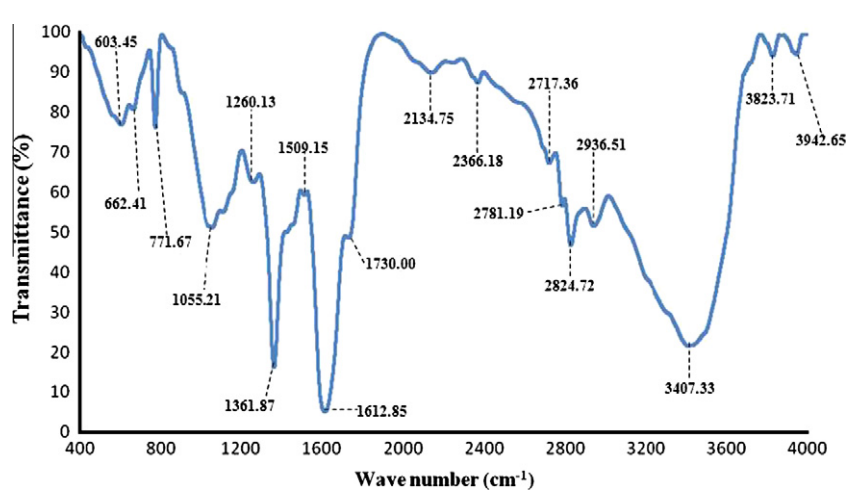

Fig. 12. FTIR spectra of SCF.

Table 3
Summary of the tensile tests (displacement rate of 0.1 mm/min; the strain rate ranged from 0.15 s^{-1} to 0.16 s^{-1}).

Gauge length (mm)	Tensile strength (MPa)	Young's modulus (GPa)	Young's modulus corrected for compliance (GPa)	Strain to failure (%)	Area (mm^2)
10	630.64 ± 46	5.09 ± 3.7	5.71 ± 5.5	12.34 ± 1.4	0.028 ± 0.003
20	656.83 ± 55	5.81 ± 2.8	6.52 ± 4.8	11.21 ± 0.9	0.023 ± 0.005
30	666.28 ± 42	6.22 ± 2.4	6.98 ± 4.2	10.54 ± 1.2	0.024 ± 0.003
40	673.12 ± 51	6.72 ± 1.9	7.54 ± 3.7	10.04 ± 1.5	0.023 ± 0.006

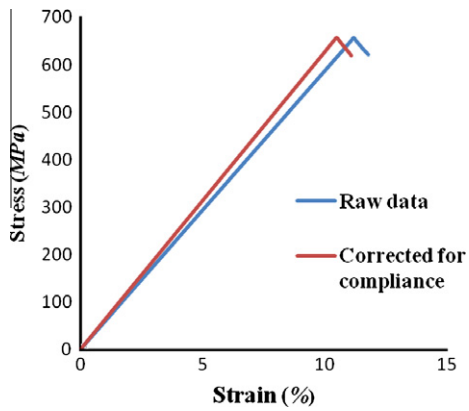

Fig. 14. Stress–strain behaviour of SCF tested at 20 mm GL showing the "as-measured" data and the data corrected for compliance.

Fig. 14 shows the "as-measured" versus the corrected stress–strain response of the SCF. At a given stress, the contribution of the machine compliance resulted in larger displacements. The modulus calculated from the corrected curve result was similar to the modulus computed from the displacement/force versus GL procedure. A non-linear region following the initial portion of the stress–strain curve was observed for some of the tested fibres. This behaviour may be due to a collapse of the weak primary cell walls and delamination between the fibre cells [19]. The strain-to-failure of fibres decreases with increasing GL. This behaviour is related to the average size and the distribution of flaws in the volume of the fibre.

The tensile strength of the fibre did not appear to be a function of GL (Table 3). The GL affects the ductility but not the strength because the mean flaw-size controls the strength. The mean flaw-size does not change with GL. When a crack is initiated at the largest flaw, the crack propagation that occurs between the flaws determines the ductility. Thus, if several flaws exist, the linkage between flaws will be quicker, and the ductility will be low (as observed in the highest GL of 40 mm). The mechanical behaviour of SCF is also dependent on its plant source, age, type of processing (mechanism of fibre extraction) and the fibre microstructure.

5. Conclusions

Four major conclusions were drawn from the test results. First, the polarised light micrograph and SEM investigations revealed that *S. cylindrica* leaves contain two types of fibres: structural fibres and arch fibres. The microstructural analysis of SCFs showed the presence of primary cell walls, secondary cell walls, fibre lumens and middle lamellae. Second, the average cross-sectional area of one of these fibres is 0.0245 mm^2. The average density and porosity fraction of the SCF is 0.915 ± 0.005 g/cm^3 and 37%, respectively. This relatively low density and high porosity fraction of the SCF are favourable for light-weight applications and in thermal and acoustic insulation. Third, XRD and FTIR analyses showed that SCFs are semi-crystalline, and I_β cellulose is present in it. Fourth, the tensile strength of the SCFs is approximately 658 MPa. The tensile strength is independent of the GL. The Young's modulus of SCFs corrected for machine compliance is approximately 7 GPa. The modulus was also not influenced by the GL. The strain-to-failure decreased from approximately 12.34–10.04% when the GL was increased from 10 mm to 40 mm.

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